"The Income component of the Human Development Index" in 2000 stated that each additional dollar of income has a smaller effect on expanding human capabilities. To ensure this effect, in calculation of GNI index natural log value is used to normalise. The log value assigns less importance to higher value compared to lower value and ensure the diminishing capability of income. For example, the natural log value of 5 is 1.60 and natural log value of 25 is 3.21. Here, the number 25 is five times higher than the number 5 but the natural log value is only two times higher. This means, the importance of higher value is getting diminished. So, the **law of diminishing capability of income** is ensured. The natural logarithm is denoted by ln or loge. The value of e is 2.718.

The sub-indices of life expectancy at birth, Education and GNI are calculated using above normalisation and the geometric mean of these three indices is worked to arrive at HDI.

Table 3.2: Sample calculation

Dimensions	Indicators	Achievement
1. Long and Healthy Life	Life expectancy at birth (years)	68.8
2. Knowledge	(i) Mean years of schooling	6.4
	(ii) Expected years of schooling	12.3
3. A decent Standard of Living	Per capita GNI (2011PPP US $)	6,353

1. Life Expectancy Index/ Health Index

$$= 68.8 - 20/85 - 20 = 0.7507$$

2. Education Index

Mean years of schooling

$$= 6.4 - 0 /15 - 0 = 0.4266$$

Expected years of schooling